

393 Final Report

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1 Introduction

This report presents the modelling and control design work for Group 13's black box system, with its interpretation as a temperature-control application. The dynamics of the temperature-control system are governed by the internal electrical system, which generates mechanical power to produce heat. The goal is to develop a linear time-invariant model from measured input-output behaviour, then to use that model to design a feedback controller that improves system performance. In particular, the controller must stabilize the power-control mechanism at the desired wattage faster than it did initially in order to meet the system targets, including a rise time of 10 seconds and a settling time of 15 seconds.

To achieve this, the black box is first examined to determine whether it can be treated as linear and time-invariant over the operating range of interest. Its noisy output is then analyzed and filtered so that the frequency-response data can be obtained more reliably. A Bode plot is used to identify key properties of the system's behaviour at different frequencies and to derive a transfer function model, and that model is then used as the basis for controller design and tuning. The motivation for this work is a temperature-control subsystem where the input is electrical power drawn by the plant, and the output is mechanical power generated, which powers the temperature-control system, allowing the application problem and the technical control problem to be combined into a single coherent framework.

2 System Application

The application considered in this report is the regulation of a temperature-control unit's mechanical subsystem serving an occupied indoor space. This choice is motivated by the fact that thermal comfort in shared environments is influenced by changing conditions such as occupancy, equipment behaviour, and other disturbances, so effective control is required rather than simple fixed operation. To improve clarity, the subsystem will be referred to as the system in the following sections.

In this context, the specific problem is the regulation of the electrical power drawn by induction motors within the system. The input is taken to be the desired electrical power drawn, and the measured output is the actual mechanical power generated by the motors. In particular, the control objective is to reduce the difference between commanded and measured power and adjust for intensive motor-startup demands so that the system behaves predictably and

can support the larger thermal objective in a controlled way.

2.1 System Assumptions

Several assumptions are required to make this formulation possible. First, it is assumed that the dynamics of the assigned black box are representative of the dominant dynamics of the induction motors within the system. Second, it is assumed that the system operates near a steady-state operating point where the physical system behaves as approximately linear. Any nonlinear dynamics can therefore be disregarded, and an LTI approximation can be used reasonably over the power range of interest. Third, although temperature regulation is the outer system motivation, power is taken as the main control variable because it is more directly measurable and manipulated within the feedback loop. The team assumed that the steady-state temperature achieved by the temperature-control system is an increasing function of the power drawn by the electrical system powering the induction motors. These assumptions allow a physically motivated but complex thermal problem to be expressed as a tractable single-input single-output control problem.

2.2 System Model

At the broader level, the application is thermal, with the desired operating condition motivating the overall system and the resulting thermal behaviour being the practical outcome of interest. At the modelling level, the focus is on the system that powers fans and compressors, which is treated as a power-regulation loop. A reference signal $r(t)$, representing the desired power draw, is compared with a feedback signal $s(t)$ to form the error $e(t)$. The controller acts on this error to generate the plant input $u(t)$, and the resulting output $y(t)$ represents the actual electrical power drawn by the system. Because the measured output contains noise, it is filtered before being fed back, which improves the usefulness of the feedback signal for control purposes.

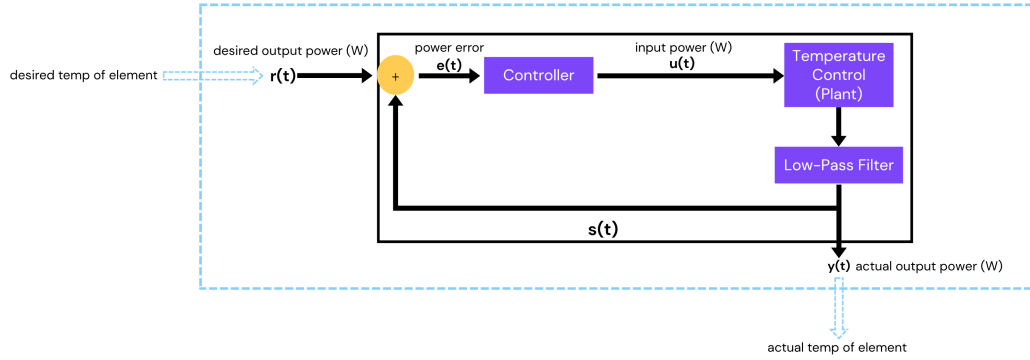


Figure 1: Closed-loop representation of the temperature-control application. The reference signal $r(t)$ represents the desired electrical power draw, while the measured output $y(t)$ represents the actual mechanical power drawn by the plant.

2.3 System Stakeholder Implications

This problem affects several stakeholder groups. Occupants depend on the system for a thermally acceptable indoor environment. Facility staff benefit from more predictable and efficient operation. Maintenance personnel are concerned with reducing overheating, electrical stress, and unnecessary faults. The institution has a financial interest in avoiding wasted energy and poor operational performance, while society more broadly is affected by the environmental consequences of inefficient energy use. For these reasons, the application is not only a suitable setting for control design, but also a problem that is practical and important. See Section 5 for a more detailed analysis of the economic, environmental, and social impacts of this problem and its stakeholders.

3 Modelling

The goal of the modelling process was to accurately construct a mathematical model for the black box system and connect its behaviour to that of the temperature-control system. First, the team clearly laid out the assumptions made about the black box system and how it relates to the temperature-control system. In order to connect the application problem of temperature regulation to the control problem of the black box system, the team assumed that the system's input and output represent the input electrical power given to

the system and the mechanical power output that powers the temperature-control system. Another key assumption was that the system was linear and time-invariant.

Given these assumptions, it is important that the team verify them through testing. Verifying these assumptions is key to developing an accurate model for the system, which is essential for the design of a controller.

Once the system had been tested and the LTI assumption was confirmed, a model could be developed. For this, the team used a zero-pole-gain transfer function model, which is appropriate under the LTI assumption. The initial parameters for the model were estimated using the Bode plot of the black box system and then tuned so that the model's Bode plot and step response closely matched the actual system's behaviour. The model developed in this section could then be used to tune a controller in the next section.

3.1 Verifying the LTI Assumption

The first characterization of the system was to assume that it was linear and time-invariant. Mathematically, this means the system $\tau : u \rightarrow y$ satisfies:

Homogeneity:

$$\tau(\alpha u) = \alpha(\tau u)$$

Superposition:

$$\tau(u_1 + u_2) = \tau(u_1) + \tau(u_2)$$

Time Invariance:

$$\tau(u(t - T)) = (\tau u)(t - T)$$

First, the homogeneity of the system was tested by entering $\sin(5t)$, then overlaying the output with $5\sin(t)$. The results are shown in Figure 2 below, where it is clear that the two graphs are similar within a reasonable error tolerance. A root mean squared error test was also done between the two outputs, with a value of 0.0444, which the team decided was sufficiently small. Within the system, if the desired power were scaled by a factor of α , it would make sense that the output would be scaled by the same amount. Intuitively, then, the system is homogeneous as well.

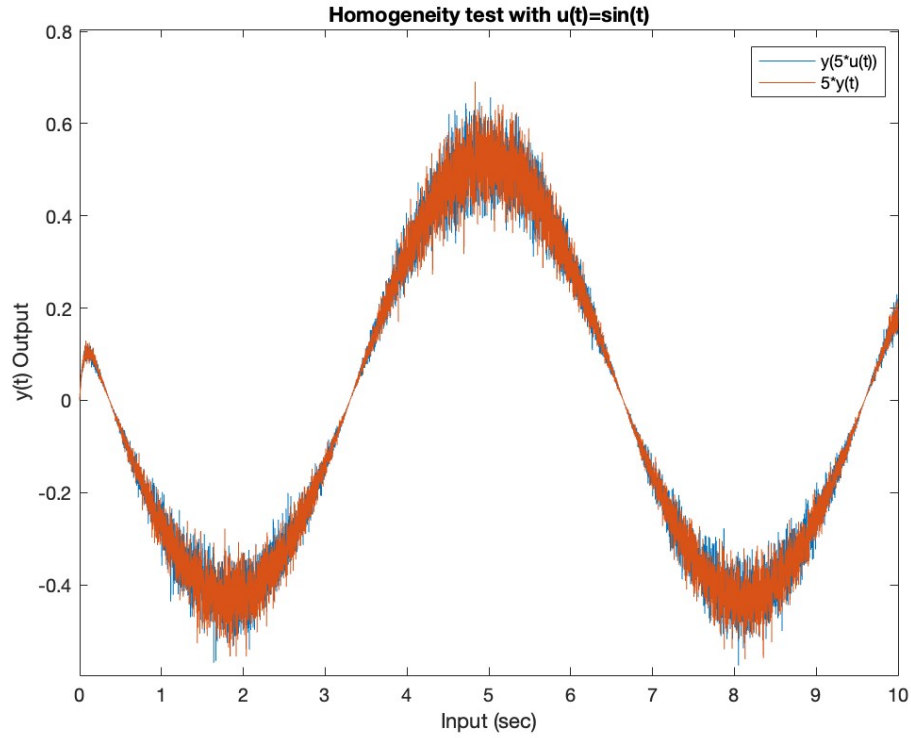


Figure 2: Homogeneity test.

Next, the system was checked for superposition. If a power of 10 Watts is input to the system, it would be expected that the resulting output is equivalent to the sum of the outputs obtained when 3 Watts and 7 Watts are input individually and then added together. Superposition was tested mathematically by running $\sin(t)$ and $\cos(t)$ separately, then adding the outputs together and plotting the result. Then $\sin(t) + \cos(t)$ was also run into the system, and the root mean squared error between the two outputs gave a value of 0.0135, which the team decided was very small. The two graphs were also plotted over each other in Figure 3, where it is clear that they are very similar within a reasonable error tolerance, demonstrating the additivity of the system.

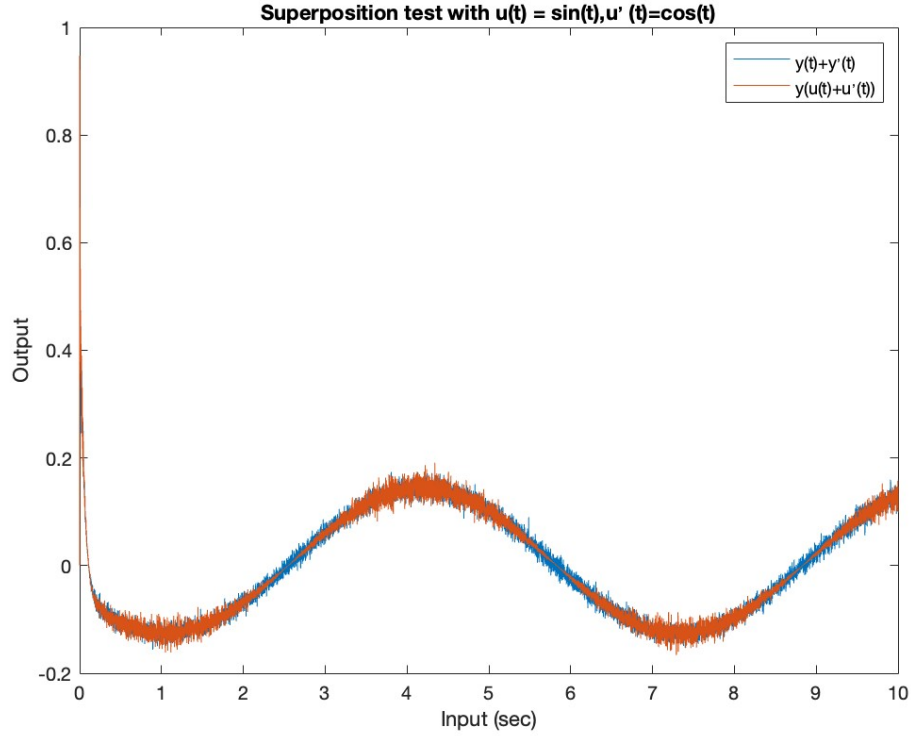


Figure 3: Superposition test.

Since the system displays homogeneity and superposition through mathematical testing as well as intuition from the application, the team concluded that it is linear.

Finally, to test for time invariance, input powers of $\sin(t - 5)$ and $\sin(t)$ shifted by +5s were run and then plotted together in Figure 4 below. The two results do not line up perfectly, but this is a consequence of the shifting of the input and the limited run time. When the plots are both active from 5 s to 10 s, it is clear that the two outputs are equal within a reasonable error tolerance. In the context of the application, there is similar behaviour, where no matter at what time the change in input power occurs, it would be expected that the response of the power drawn would be the same when shifted for the time difference. This shows the time-invariance property of the system.

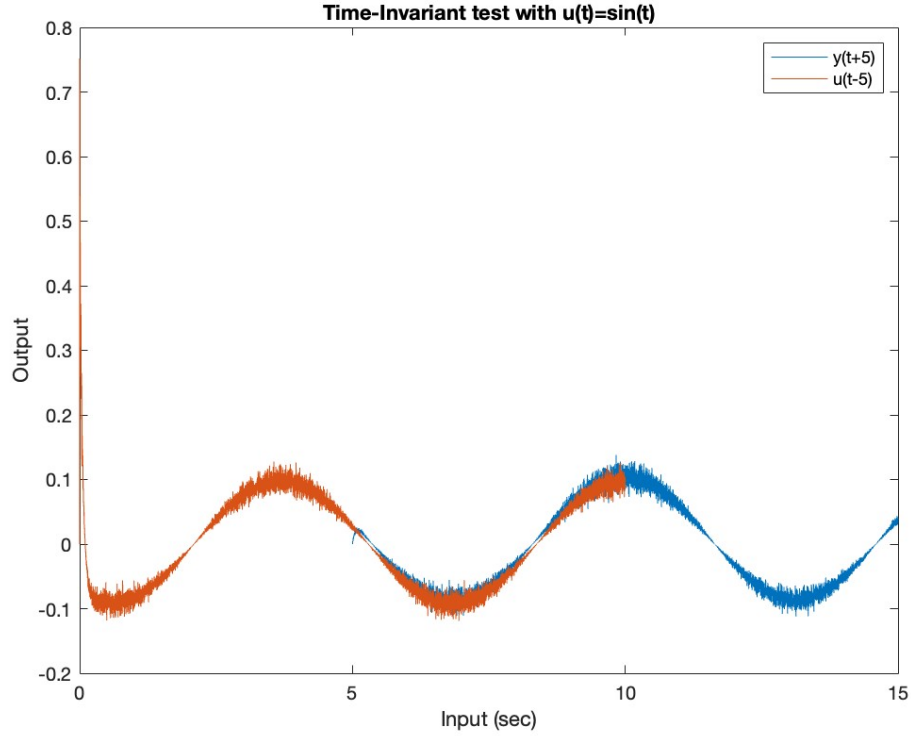


Figure 4: Time invariance test.

From all the quantitative testing of the behaviour of the system and the intuitive testing of the behaviour of the application, the group felt justified in assuming that the system possesses linear time-invariant properties. This is a very useful property because it makes the modelling of the system easier, exhibiting intuitive properties that make the model simpler to understand. Linear time invariance also allows more specific system analysis, such as generating a Bode plot.

3.2 Noise Classification and Filtering

Now that the LTI properties of the system were verified, it was time to identify and characterize the noise of the system. The first step was to run the same power input $\sin(t)$ 500 times, then average the power drawn at each data point over all trials. This produced Figure 5 below, which shows an ideal noiseless version of the output power.

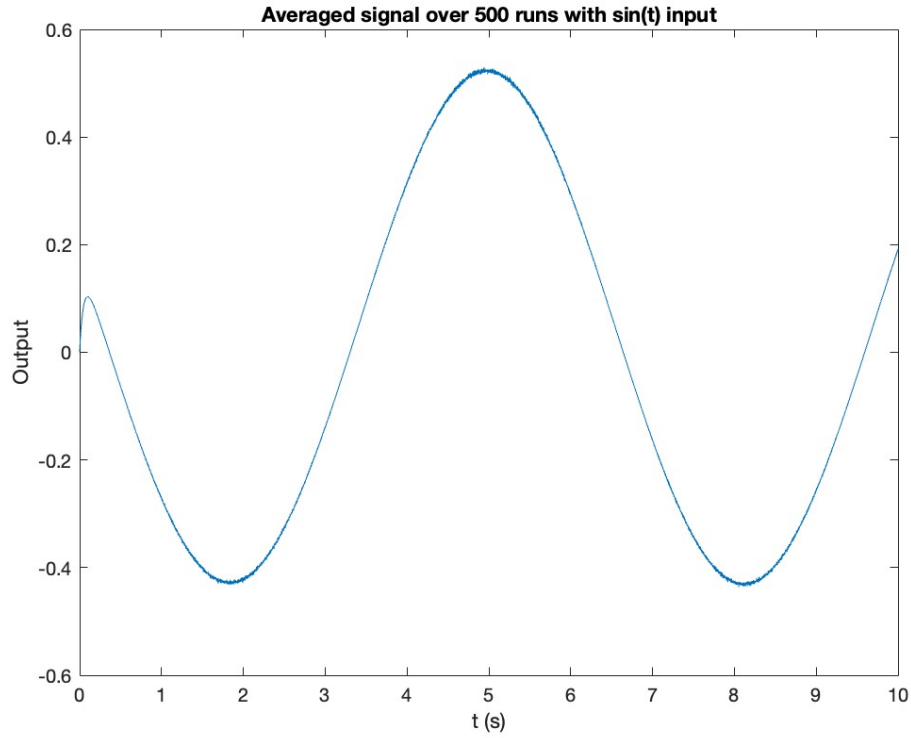


Figure 5: Average of 500 power inputs.

The noiseless output was then subtracted from a randomly chosen simulation of the 500, generating a graph of the noise as seen in Figure 6. The correlation between the averaged output and the noise was calculated to be 0.534, which indicates a positive non-zero correlation. The team also observed that the amplitude of the noise grew and shrank with the amplitude of the system's output, implying multiplicative noise.

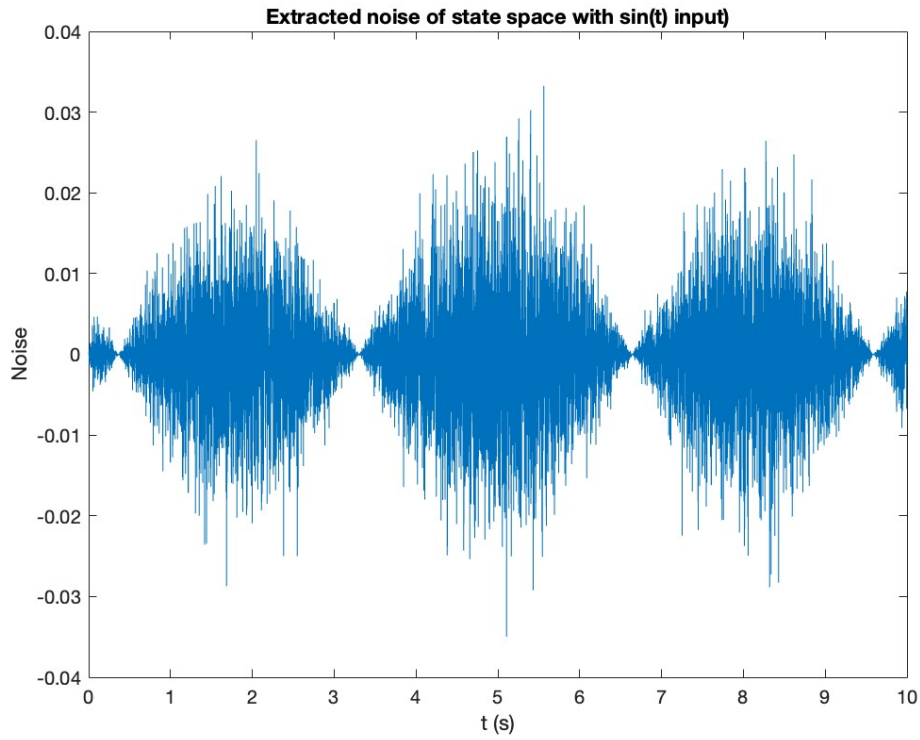


Figure 6: Plot showing extracted noise by taking the difference between the average and a random simulation.

Moreover, these values were plotted on a histogram, which sorted the noise based on the occurrence of each amplitude. This resulted in a sharp bell-curve shape, leading the team to conclude that the system exhibits Gaussian noise.

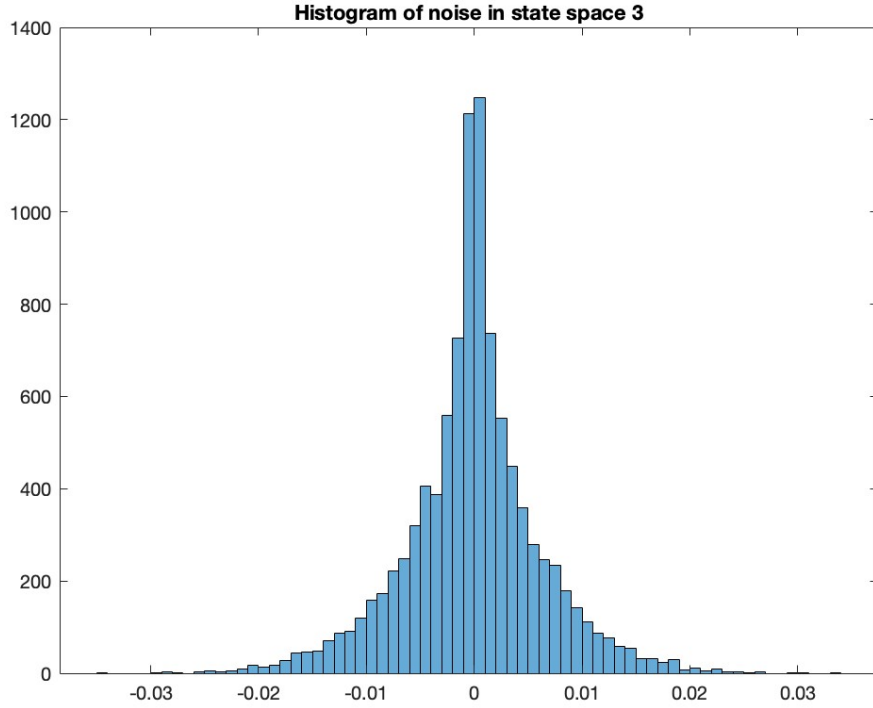


Figure 7: Histogram of extracted noise values.

Finally, using Welch's power spectral density estimate, the power of each normalized frequency of the noise was estimated, creating Figure 8. The power spectral density has a lot of noise, but the values hover around -50 dB/rad/sample over all normalized frequencies. This led the team to determine that the system's noise had a component of white noise.

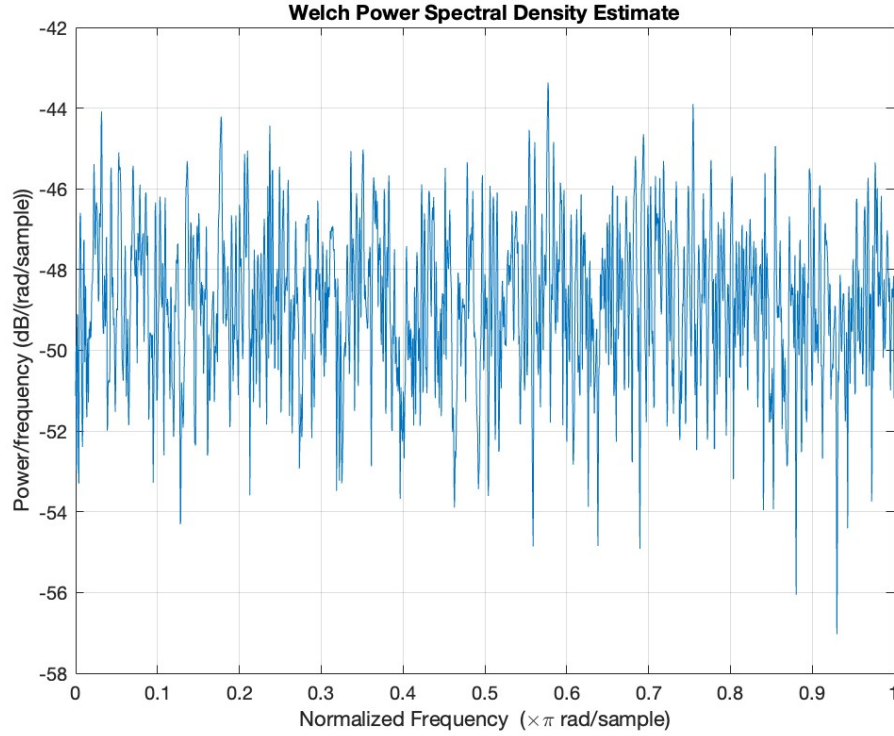


Figure 8: Power spectral density graph of system noise.

The noise has been identified as multiplicative Gaussian noise with an element of white noise. In the context of the application, this noise could be a product of inefficiencies within the measurement tools, the random motion of free electrons due to the finite temperature of the components (thermal noise), or inefficiencies in the circuitry [1]. It makes sense that the noise would be multiplicative since, as the power of the system increases, the thermal noise would increase as well. The presence of white noise is also intuitive since that would likely stem from circuit and measurement noise, which are independent of frequency.

Now that the noise has been identified, it needs to be filtered. This was done by applying multiple filters to the power drawn and observing the result compared to the original output. The team found that the best result came from using a low-pass filter with a cutoff frequency of 0.001 Hz; the results can be seen in Figure 9. The root mean squared error between the averaged ideal power output and the filtered output was ± 0.0016 , which was the lowest value achieved across all the different filters and cutoff frequencies applied. The main role of such a filter is to ensure that spikes in power drawn from the

system do not throw the controller into chaos. The system is focused on the overall power drawn over time rather than the instantaneous power output. It is important to note how the filter changes the overall amplitude and thus the gain, meaning that for practical use an amplifier may need to be applied at some point.

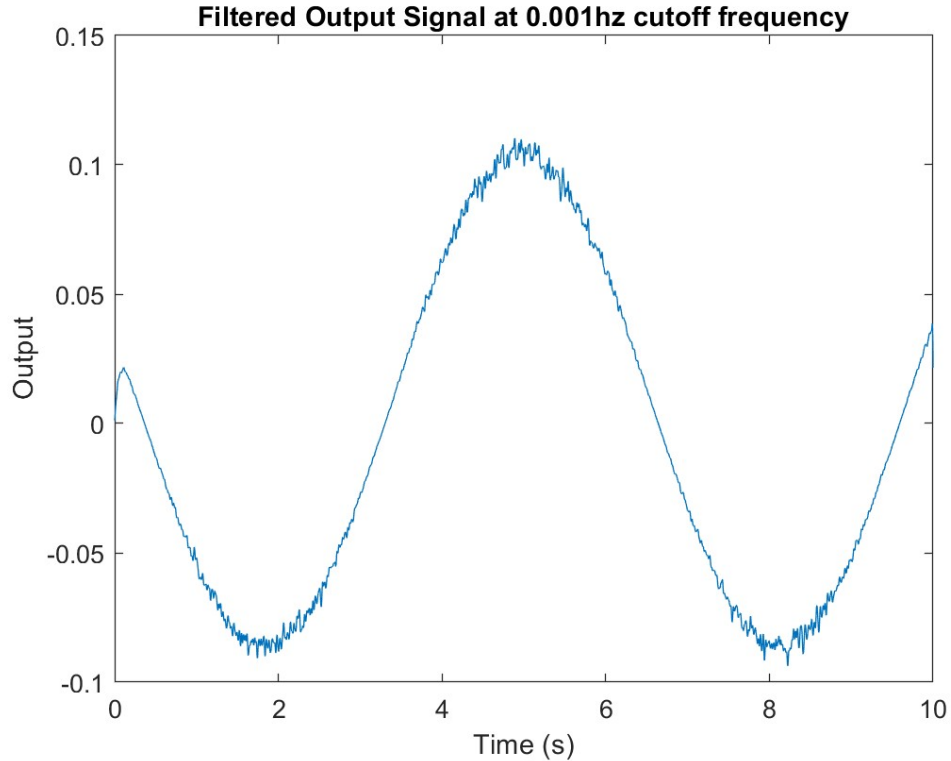


Figure 9: Output of the system with $\sin(t)$ input and a low-pass filter applied to the power drawn.

3.3 Constructing the Bode Plot

The first step in constructing a model was to generate the Bode plot for the system in order to examine its frequency response. This was done by applying a range of pure frequencies as input to the system in the form $u(t) = \sin(2\pi ft)$ where f ranges from 10^{-4} to 10^4 Hz. Following from the presumed LTI property of the system, the output is also periodic with the same frequency, and in particular the output is of the form $y(t) = G \sin(2\pi ft + \phi)$ where G is the magnitude or gain for that frequency and ϕ is the phase shift. One example of an input applied to the system is found below.

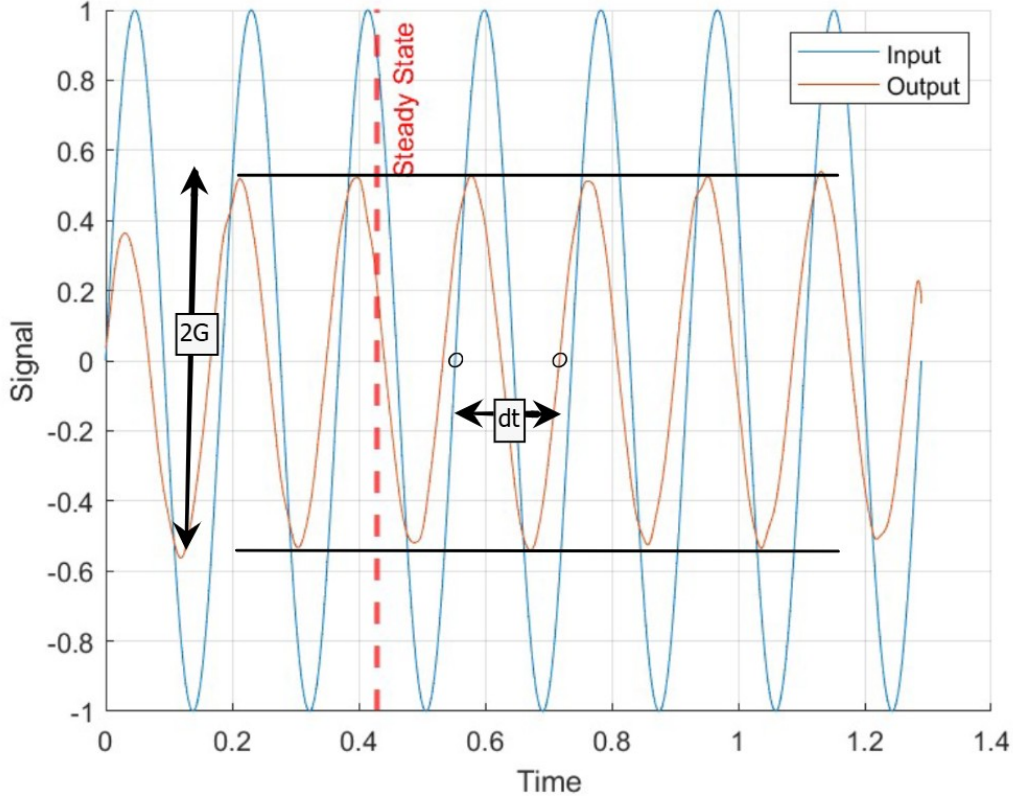


Figure 10: Output of the system with a periodic input, with Δt and G shown.

In the figure above, the blue graph shows the input, and the orange graph shows the output of the system with a shifted phase and decreased magnitude. In order to capture the steady-state response of the system, the initial data points of each plot are not used to find the gain or phase shift, and this steady-state point is indicated by the dotted red line in the plots. The value G in the figure above represents the absolute gain of the system, $G = \frac{A_{out}}{A_{in}}$, where A_{in} and A_{out} are the amplitudes of the input and output waves respectively. In all samples, $A_{in} = 1$, so this is simply the amplitude of the output wave, which was found by measuring the maximum and minimum values attained and dividing the difference by 2. Then the gain (in decibels) of the output for any given input is given by

$$G_{dB} = 20 \log_{10}(G) \quad (1)$$

This is the quantity plotted in the Bode plot.

The phase shift was measured by first finding the value labelled ' Δt ' in the above plot and given by $\Delta t = t_1 - t_0$, where t_0 is the first time after steady

state when the input signal hits 0 while increasing. t_1 is the first time after t_0 when the output signal hits 0 while increasing. The phase shift (in degrees) is then calculated as $\phi = 360f\Delta t$, with f being the input frequency. Plotting the gain and phase against frequency on a log scale gives the Bode plots shown below.

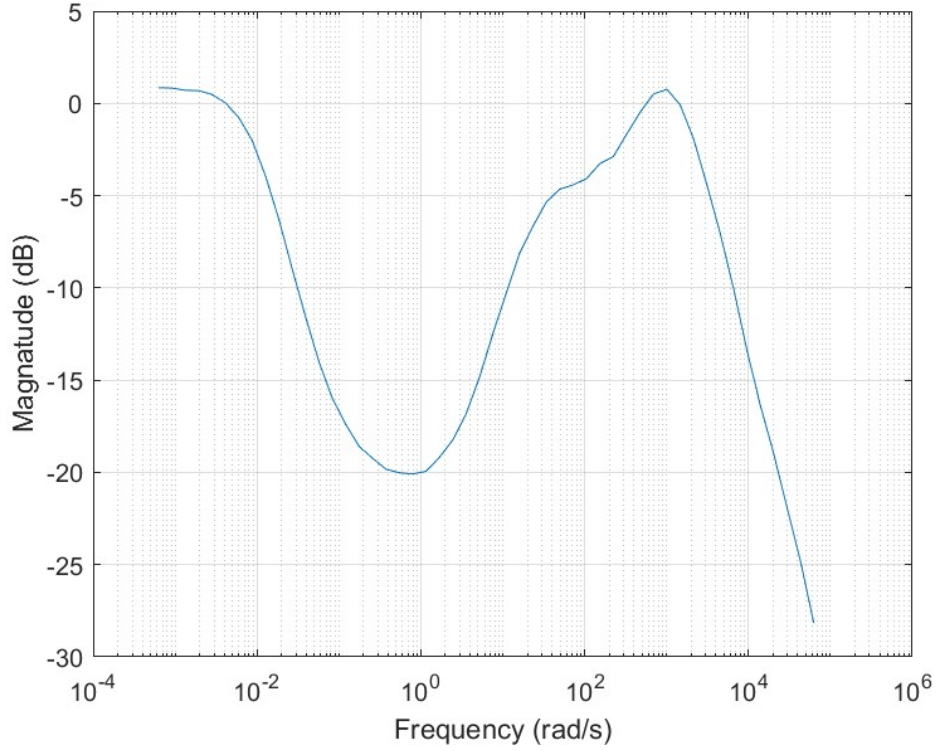


Figure 11: Plot of G_{dB} against f .

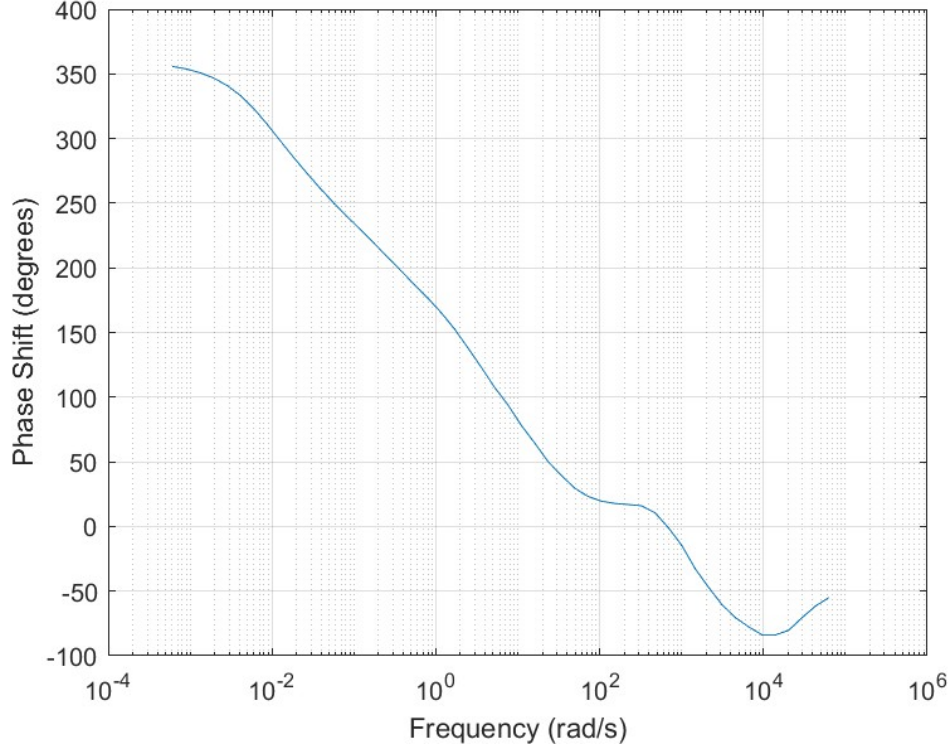


Figure 12: Plot of ϕ , the phase shift, against the frequency input f .

3.4 Estimating and Fitting the Model

Given that the system is assumed to be LTI, it is modelled with a zero-pole-gain transfer function of the form

$$H(s) = k \frac{(s - z_1) \cdots (s - z_n)}{(s - p_1) \cdots (s - p_m)} \quad (2)$$

where $m > n$ and all zeros and poles are real-valued. The initial estimates of these values were obtained from the Bode plot in the previous section. The team observed that the slope of this plot is roughly constant over certain intervals. Places where the slope changes by ± 20 dB/decade are labelled as approximate zeros and poles of the transfer function. This process is shown in the plot below.

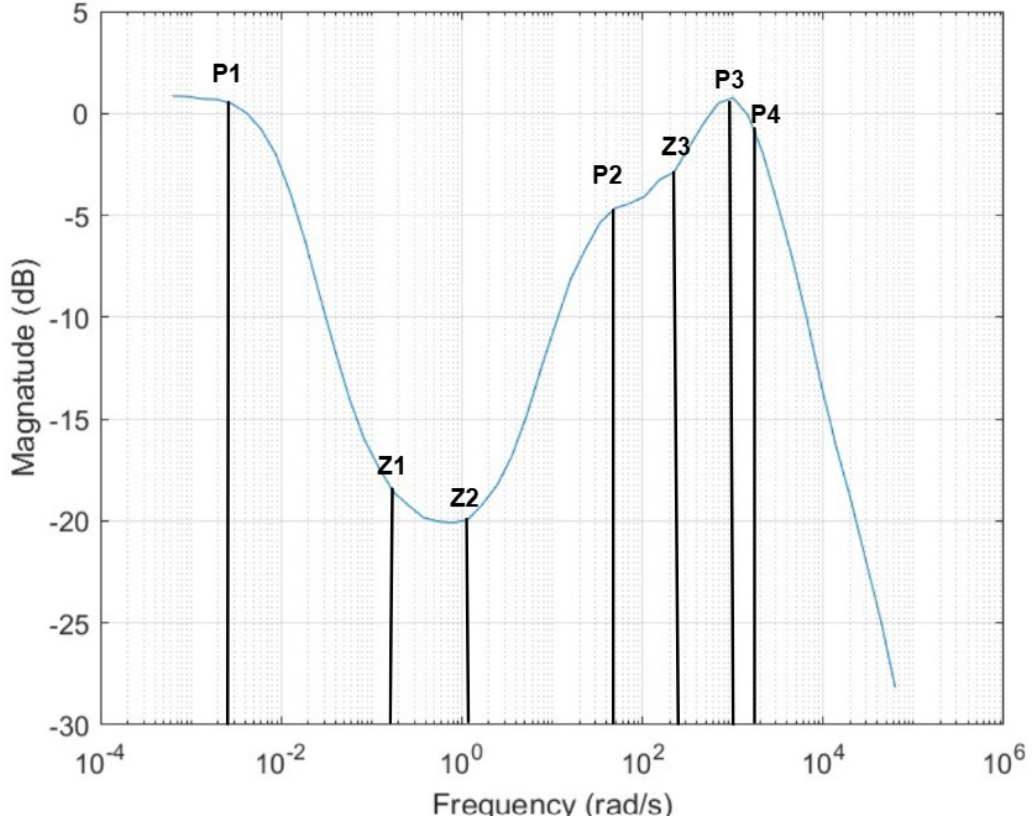


Figure 13: Plot showing the initial approximations for the zeros and poles of the model.

From the above plot, the team estimated that there are three zeros and four poles governing the behaviour of the system and labelled them $(z_1, z_2, z_3) = (0.1, 1, 115)$ and $(p_1, p_2, p_3, p_4) = (0.002, 50, 1000, 1950)$. The initial estimate for the parameter k can then be calculated using the value M_0 , which is the DC magnitude value achieved by the system when a low frequency is applied. The team estimated this value to be 1.20 from the Bode plot. The team then applied the transfer function formula

$$G(0) = 10^{\frac{M_0}{20}} = k \frac{(z_1 z_2 z_3)}{p_1 p_2 p_3 p_4} \iff \frac{p_1 p_2 p_3 p_4}{(z_1 z_2 z_3)} G(0) = k \quad (3)$$

where $G(0)$ is the absolute gain. Using this formula with the estimated poles and zeros, the team obtained $k = 19469$. The transfer function specified with these parameters did not behave similarly enough to the model, so manual parameter tuning was required to fit the parameters correctly. Namely, the

initial gain was far too high, so it was lowered by an order of magnitude. From here, the goal was to fine-tune the model so that its unit-step response closely resembled that of the original system. After manually tuning the parameters to approximately match the behaviour, the MATLAB `tfest` function was used to fit the model based on the current rough estimate and the step response of the actual system. The equation for the transfer function is given by

$$G(s) = 2850 \cdot \frac{(s + 461)(s - 3)(s - 0.1)}{(s + 2598)(s + 842)(s + 20.3)(s + 0.01)}$$

This model was able to achieve the desired behaviour as shown below.

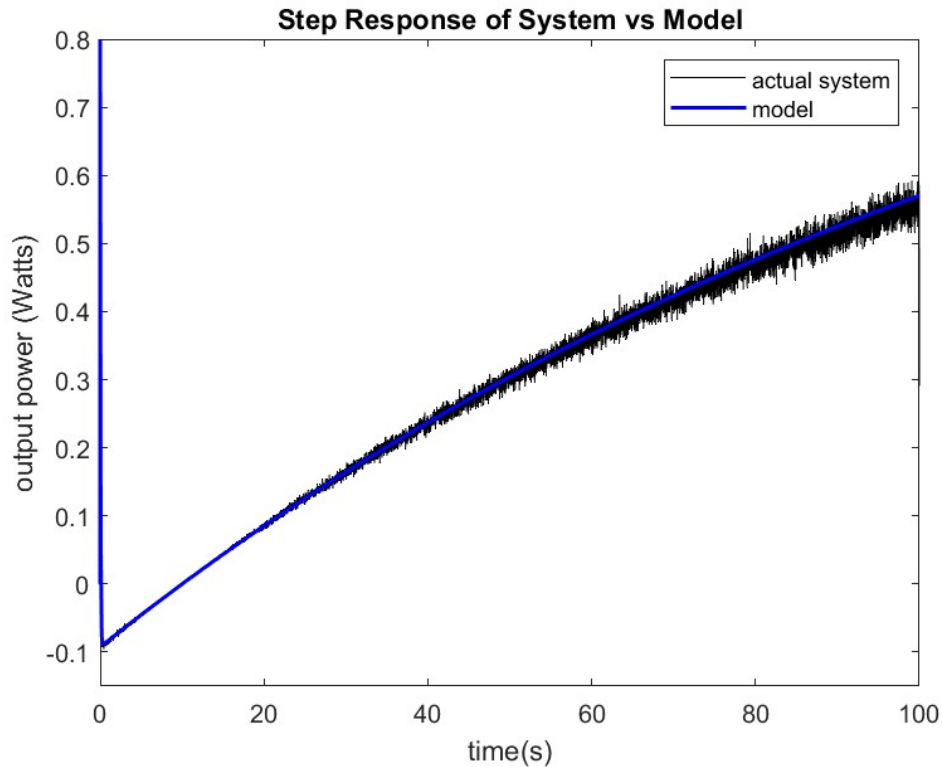


Figure 14: Plot showing the step response of the model compared with the actual system.

The model's behaviour is very similar to that of the system. As a second indicator that the model is accurate, the team plotted the Bode diagram for the model.

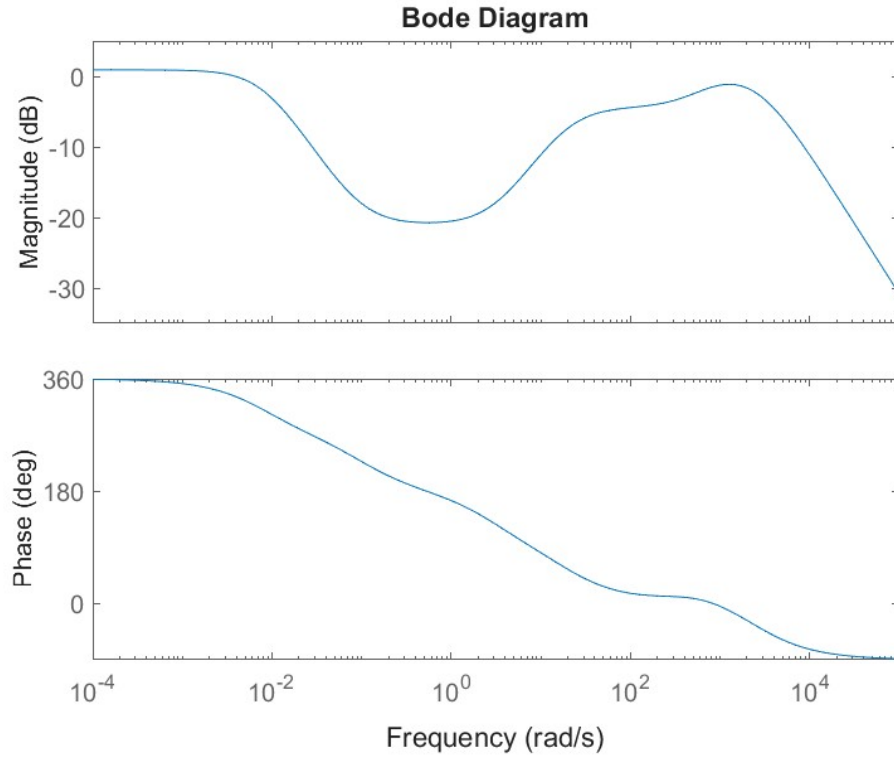


Figure 15: The Bode plot of the model.

The Bode diagram closely resembles the one generated from the actual system, reinforcing the validity of the model.

4 Control Design

The next stage of the project was to develop a controller based on the model developed in the previous section. The goal of the controller was to increase the speed at which the system stabilizes and to maintain the desired output when a constant input is applied while minimizing overshoot. Over the course of the project, the team focused on building a PID controller for the system because of its general robustness and ease of implementation. In addition, it is a controller type widely used in engineering applications, giving a useful baseline for what to expect and many examples against which the system can be compared. Using the model derived in the previous section and constraints based on the system requirements, the controller was developed for the system. Using the PID Tuner and manual adjustments of parameters,

the group created a transfer function model of the controller which was able to meet most of the system requirements and best suited the system application.

4.1 Final PID Controller

Initially, the controller was designed and tuned using the PID Tuner app in MATLAB and based on the model derived in Section 3. Through this modelling, the team was able to design a controller which met the system requirements and stabilized the output signal while using the model transfer function as the plant. The reason for using the model rather than the actual black box system to initially tune the controller is that it is much faster to compute the output signals based on the numerically defined model using the PID Tuner interface than it is to run a signal through the black box system directly. When applying the controller to the actual black box system, the team found that it did not quite meet the required specifications, and so the controller was further tuned by manually altering controller parameters while using the actual black box system as the plant.

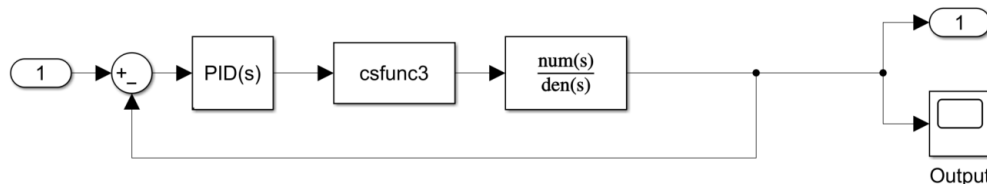


Figure 16: Figure showing the final Simulink model used.

In the end, the group settled on the controller given by:

$$C(s) = 7.5 + \frac{0.075}{s} + 0.125 \cdot \frac{100s}{s + 100} \quad (4)$$

Throughout the manual tuning process, the group found a great deal of success in reducing the proportional and integral values to reduce the system's undershoot at the cost of stabilization speed. The team also used a low-pass filter ($\text{num}(s)/\text{den}(s)$ in Figure 8) with a cutoff frequency of 30 Hz to best filter out noise. More aggressive filters were tested; however, the overall gain of the system was affected, causing larger unwanted changes to the system's control, and in some cases would create a pole in the system. Other controllers were

considered, particularly PI control models, which were found to be successful and appeared in preliminary application research for induction motor control.

4.2 Controller Performance

The black box system came with performance specifications that the controller was required to meet:

Table 1: System target values and recorded values

	Steady State Error	Rise Time	Overshoot(%)	Settling (s)	Settling (ϵ)
Targets	± 0.08	10	1.4	15	0.08
Controller	-0.04	12.5	0	13	0.09

As seen above, the controller meets the specifications for steady-state error, overshoot, and settling time, but does not achieve the rise-time target or the settling-error target. This trade-off was made in order to reduce the system's undershoot and better account for power overdraw, which is common in induction machines at startup. This can be seen in the following step response of the system:

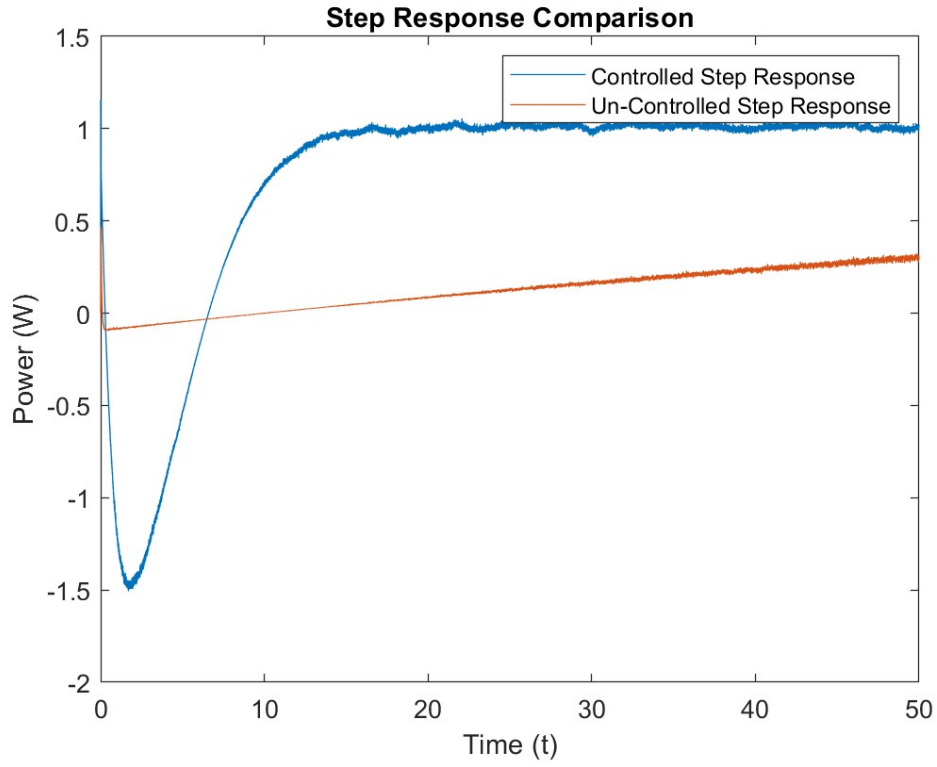


Figure 17: Step response of the controlled and uncontrolled system.

It is important to note that negative power implies a change in direction of the motor's rotor, and not a generation of power. Keeping this in mind, the importance of reducing the undershoot is to reduce the overall physical strain caused by aggressive direction switching in the system in order to increase motor life.

In addition to a step response, Group 13 also performed a simulation to model the response in the context of a three-phase power system used for induction motors with the equation:

$$r(t) = (400e^{-t/5} + 200) \left[\sin^2(120\pi t) + \sin^2\left(120\pi \left(t + \frac{2\pi}{3}\right)\right) + \sin^2\left(120\pi \left(t + \frac{4\pi}{3}\right)\right) \right] \quad (5)$$

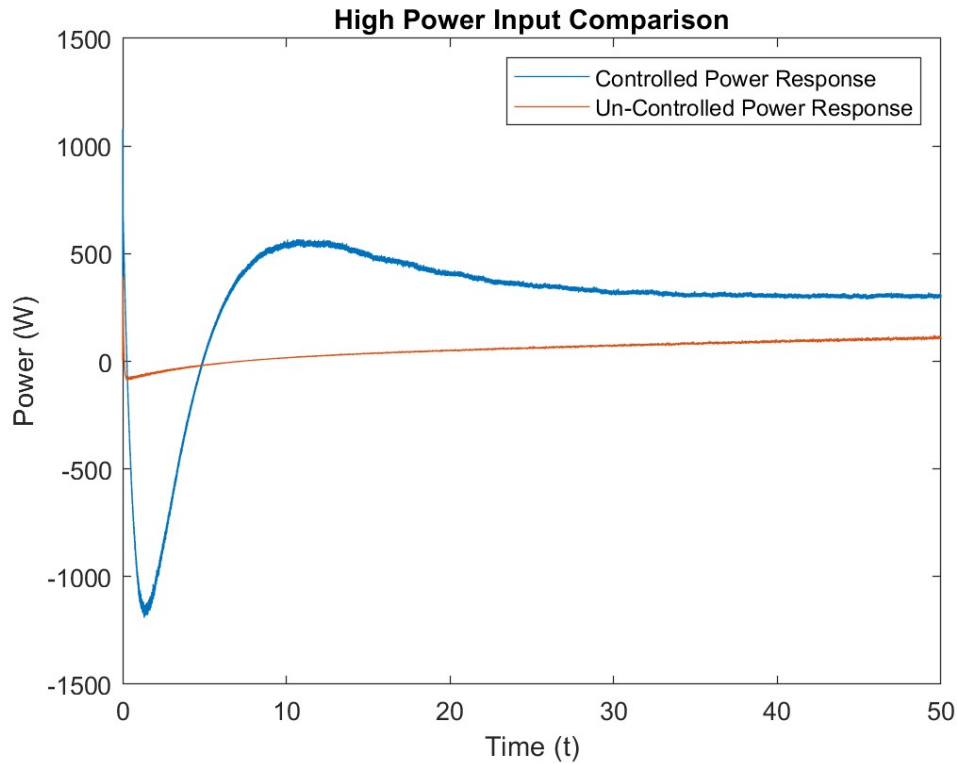


Figure 18: Response of the controlled and uncontrolled systems with 3-phase decaying power.

Similarly to the step response, the controlled system reaches the ideal power, in comparison to the uncontrolled system, which lags behind. In addition, while the power draw at the undershoot is incredibly heavy, induction motors can generally sustain above 6-7 times rated power for brief time periods to account for startup power[2], as such the temporary heavy draw is acceptable.

5 Economic, Environmental, and Social Impacts

In order to perform a detailed triple bottom line analysis, the team chose to evaluate the system's impacts in terms of its application to learning spaces on Queen's University campus.

5.1 Economic Evaluation

The economic value of the proposed energy-regulation system becomes evident when examining the daily energy wasted in a single lecture hall. Based on Ontario Energy Board pricing, running 1 kW over a typical 10-hour academic day costs approximately \$0.1754 [3]. In a standard one-hour class with roughly 120 occupants, the additional heat load from people alone contributes an estimated 10–15 kW per hour [4]. Using a midpoint of 12 kW across an average of six classes per day results in roughly 72 kWh of wasted energy. Applying the previously calculated cost rate yields $72 \times 0.1754 = \$12.63$ in avoidable energy expenses per day. These figures show that even a single room experiences measurable unnecessary costs, underlining the financial value of implementing a system that actively regulates HVAC power draw.

5.2 Environmental Evaluation

Queen’s 2024 Carbon Footprint Report emphasizes that regulating HVAC power draw has significant environmental benefit, as currently over 77% of all campus emissions come from natural gas-based heating systems [5]. With regard to the team’s system application, existing retrofits demonstrate that even small reductions in heating demand translate directly to meaningful carbon reductions; the LaSalle Building’s 2024 ventilation recirculation upgrade, for example, produced an 80% decrease in heating energy use, saving 75 t CO_2e /year [5]. By implementing a system that regulates the electrical power drawn by lecture hall temperature-control systems, the team’s solution targets the same high-impact category of emissions without requiring invasive renovations. Such renovations are not viable considering the age and history of many buildings at Queen’s.

The Climate Action Plan also stresses that “upgrading building controls, including scheduling and occupancy-based HVAC, will be a primary focus moving forward...” [5]. The team’s proposed embedded hardware control system directly aligns with this strategy by minimizing unnecessary heating and cooling loads in real time. Because the system adjusts power draw to match a desired setpoint, it reduces strain on aging equipment, lowers natural gas consumption, and decreases electricity use during periods when Ontario’s grid is more carbon intensive.

Lastly, many of Queen’s older buildings rely on aging HVAC systems, which present a higher fire risk due to worn electrical components, overheating motors, and inconsistent power draw. When these systems operate inefficiently or under fluctuating loads, they degrade faster, leading to premature disposal and

increased landfill waste. While full HVAC replacement or major renovations could address these risks, they are costly, disruptive, and often unnecessary if the equipment still functions. The team’s power-regulation system stabilizes electrical demand, reduces thermal stress on components, and extends equipment lifespan, offering a safer and more sustainable alternative to large-scale mechanical overhauls.

5.3 Social Evaluation

In a typical lecture hall, occupancy patterns cause significant temperature fluctuation. Similarly to what was stated in the economic evaluation, with 5 to 7 classes per day at roughly 60% capacity, and each person emitting about 100 watts of heat, the room’s temperature can rise from 22°C to 27°C within an hour [6] [7]. Not only does this create a health concern for those vulnerable to overheating, but it also negatively affects the learning abilities of students. According to the European Commission, even a 1°C increase above the ideal learning temperature can reduce cognitive performance by about 2 percent [8]. By regulating HVAC power draw every ten seconds, the team’s system prevents this heat buildup and keeps conditions stable, improving the comfort, focus, and safety of students and staff.

For maintenance staff, the system reduces strain on aging HVAC units and lowers the likelihood of dangerous overheating events. Older systems, especially those 10–15 years old, are more vulnerable to electrical stress and can become fire hazards if overworked [9]. Installing the new hardware creates an opportunity to assess equipment condition and make repairs. This not only improves workplace safety for maintenance personnel, but also reduces emergency repair demands and extends system lifespan, all without the disruption or cost of major building renovations.

5.4 Regulatory Compliance

An important consideration for this application is whether the proposed controller operates in a manner consistent with recognized electrical engineering practice for AC induction motors. For industrial and commercial power systems, the most relevant references are IEEE Std 3002.7-2018 on motor-starting studies and IEEE Std 3004.8-2016 on motor protection [10, 11]. These recommended practices frame acceptable motor-starting behaviour in terms of the full starting condition, including transient current, associated voltage drop, and protection of the motor during starting, rather than by a single universal

limit on startup overdraw alone. This is consistent with standard induction-motor behaviour: locked-rotor current is typically about 600–700% of full-load current, and direct-on-line starting commonly produces currents of about 6–8 times rated current for a short duration [12, 13].

In the context of this report, the plant is interpreted as an induction-motor-driven temperature-control subsystem, and the controller was tuned with startup demands in mind. The controlled response was designed to reduce aggressive transient behaviour and avoid unnecessarily severe reversals in the mechanical system during startup. From a practical protection standpoint, this is a reasonable design goal, since repeated or severe starts subject motor windings, insulation, and rotating components to additional thermal and mechanical stress [12]. The system therefore behaves in a way that is directionally consistent with IEEE-recommended practice, in the sense that startup overdraw is treated as expected but something that should still be moderated and managed.

At the same time, the present results are not sufficient to claim formal IEEE compliance in a strict design-verification sense. The current report gives closed-loop power-response metrics, including a rise time of 12.5 s and a settling time of 13 s, but it does not provide a formal motor-starting study showing line current, terminal-voltage drop, overload or relay coordination, or a thermal withstand check for the motor and associated protection devices. In addition, the reported rise time is a control-performance metric for the power response of the modelled system, not necessarily the same as the motor acceleration time used in industrial starting studies. For this reason, the most accurate conclusion is that the proposed controller appears broadly consistent with IEEE-recommended motor-starting and protection practice for AC motors, but full compliance would require additional installation-specific verification using the methods described in IEEE Std 3002.7-2018 and IEEE Std 3004.8-2016 [10, 11].

6 Summary and Future Improvements

6.1 Report Summary

Group 13 began with a black box system and interpreted the unknown dynamics of the system as a temperature-control application for indoor spaces, with the goal of controlling the power drawn by the temperature-control system to a desired power level faster and more reliably. The input of this system is the desired power, and the output is the actual power response of the sys-

tem.

The team assumed the black box can be treated as a linear time-invariant system, and this assumption was justified by testing homogeneity, superposition, and time invariance. These properties were tested mathematically, graphically, and intuitively with the application, leading the team to conclude that the system behaves closely enough to linear and time-invariant for modelling purposes. The noise created by the system was then extracted and tested in order to determine that the noise is mainly multiplicative Gaussian noise with a white-noise component. In order to filter out this noise, the team found that a low-pass filter with a 0.001Hz cutoff gives the best match to the averaged “ideal” output. The team then constructed a Bode plot by inputting sine waves over a wide range of frequencies and measuring output gain and phase shift. This reveals how the system responds at different frequencies, allowing the team to estimate a zero-pole-gain model with three zeros and four poles. Rough estimates were first made by observing slope changes in the bode plot, then the parameters were tuned manually and with MATLAB’s `ttest` function. The team was then able to find a transfer function that matched the step-response and bode plot of the actual system closely enough to use for controller design.

The group chose a PID controller because it is common, robust, and easy to implement, first tuning it in MATLAB using the model, then adjusting it further using the real black box. The final controller improves system performance and meets most target requirements. It achieves acceptable steady-state error, zero overshoot, and settling time, but misses the rise-time target and slightly misses the settling-error target. These missed targets were accepted by the team to reduce undershoot and aggressive startup behaviour, because large reversals or overdraw can strain induction motors and shorten motor life. The controller was also tested on a simulated three-phase decaying power signal meant to represent induction motor conditions and the controlled system follows the desired behaviour much better than the uncontrolled one.

A triple bottom line analysis of the system was also done, focusing on lecture halls at Queen’s University. Environmentally, the controller could help reduce emissions by lowering unnecessary HVAC demand, especially since campus heating emissions are a major source of carbon output. The life of the equipment could also be extended by not straining the system, reducing waste from early replacement. Socially, the system could improve comfort and learning conditions in classrooms by reducing temperature swings. The maintenance staff will also have improved safety by reducing strain and overheating in aging HVAC equipment. When it comes to regulatory compliance, the controller seems consistent with recommended practice, but the group notes that full

compliance cannot be claimed without more detailed motor-starting and protection studies.

6.2 Future Improvements

Given more time to work on the project, and fewer constraints to consider, the group would have done some of the following. First, the team would refine the model. Generally, motor systems are described using voltage and current as inputs, and torque and angular velocity as outputs. Turning the system into a multivariable problem would allow the team to be more specific about what needs to be controlled and regulated with regard to system demands and to be more in line with standardized regulations for motor and temperature-control systems. The group would also build out models for other subsystems within the overall temperature-control system, such as boilers, compressors, and sensors, to provide a stronger proof of concept for implementation of the system and to test with respect to disturbances that may be caused by internal issues.

In addition, the group would perform benchmarking of the system in comparison to industry and academic standard models used for motor and HVAC systems to compare the stability and performance of the model. Part of this work would include non-ideal behaviour within the system, such as losses due to machine inefficiencies, testing with noisy inputs, and implementing some nonlinear behaviour found within induction motors. This would give the team a stronger idea of where the system lies in terms of performance and standards, and whether the system can truly be assumed to be a subsystem of a temperature regulator.

Moreover, while the controller generally met specifications within the context of the system, not all system targets were fully met, and there is still room to improve the system. In particular, given more time, the group would have continued tuning the controller to reach a better balance between application and system requirements, improving certain aspects such as rise time and undershoot. In addition, the team would explore other control models other than a standard PID feedback system, such as a feed-forward method proposed in Week 12's class, or models that have been proposed for motors such as General Predictive Control [14] and Fuzzy PID Control [15], to see if they better suit the black box system and the application.

The team believes these future steps would address limitations and issues found within the system, create a stronger conceptual model, and improve the overall performance of the system.

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